Supplementary: Disparity analyses are robust to ancestral state estimation uncertainty

Caleb Scutt[1,2*], Natalie Cooper[2], Gavin Thomas[1], Thomas Guillerme[1]

[1]School of Biosciences, University of Sheffield, Sheffield, UK

[2]Science Group, Natural History Museum, London, UK

*Corresponding author: cnscutt1@sheffield.ac.uk

S1 Supplementary Tables

Available on Dryad at `https://doi.org/10.5061/dryad.280gb5n4p`.

Table S1 `/continuous/continuous_aggregated_method_multcomp.csv`

Table S2 `/continuous/continuous_aggregated_method_model_fossil_multcomp.csv`

Table S3 `/discrete/discrete_aggregated_method_multcomp.csv`

Table S4 `/discrete/discrete_aggregated_method_model_fossil_multcomp.csv`

S2 Supplementary Figures

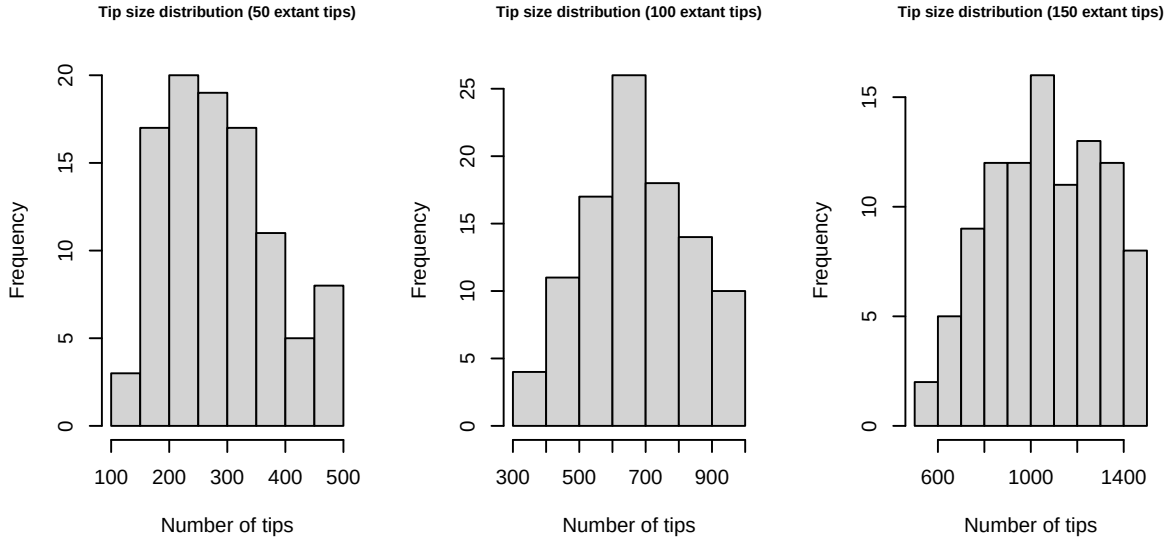

Figure S1: **Distribution of simulated tree sizes.** Histograms show the final number of extant tips across the three targeted size regimes (50, 100, and 150 extant taxa). Minor variations around the target sizes reflect the stochastic nature of the birth-death simulation process.

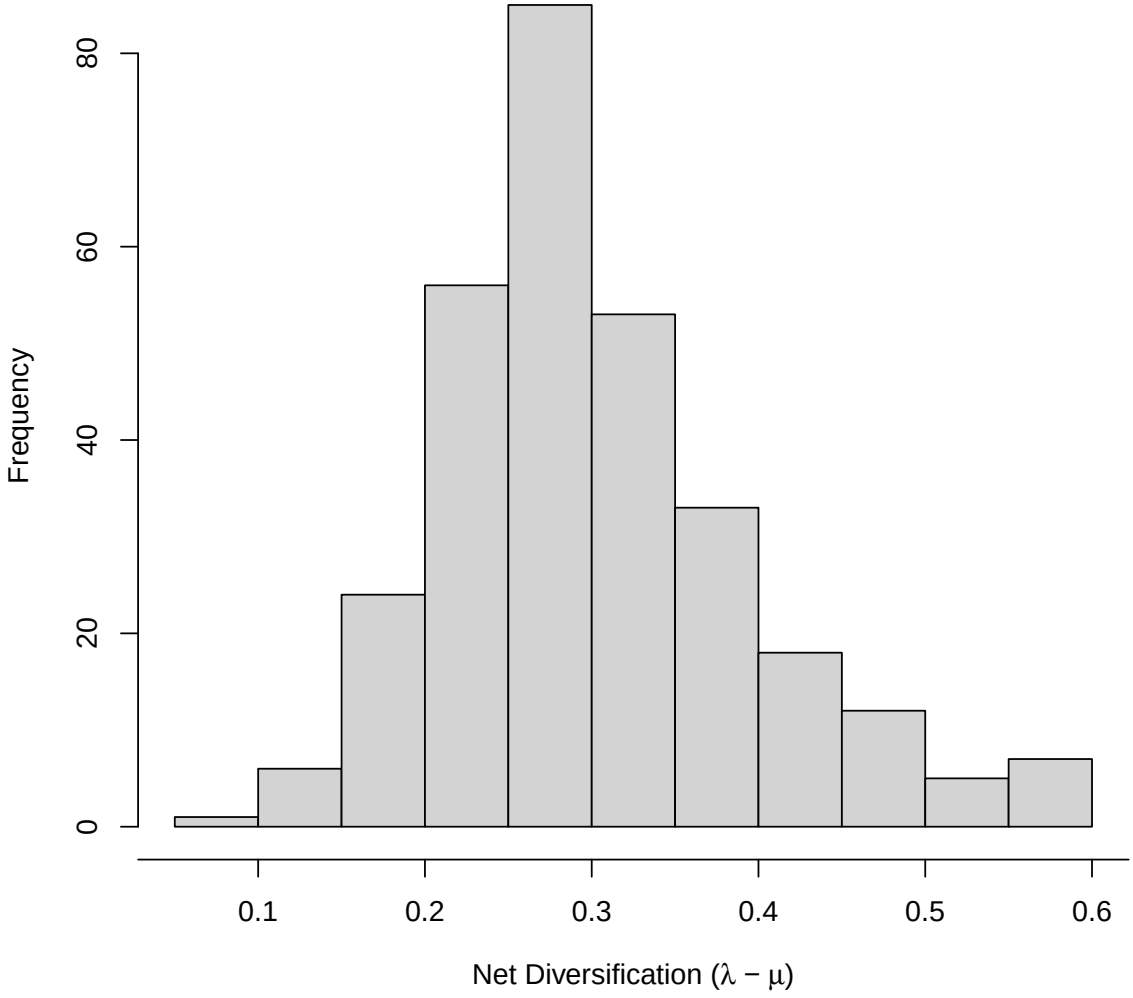

Figure S2: **Birth-death parameters for tree simulation.** The distribution of net diversification (speciation minus extinction) values across all simulated phylogenies.

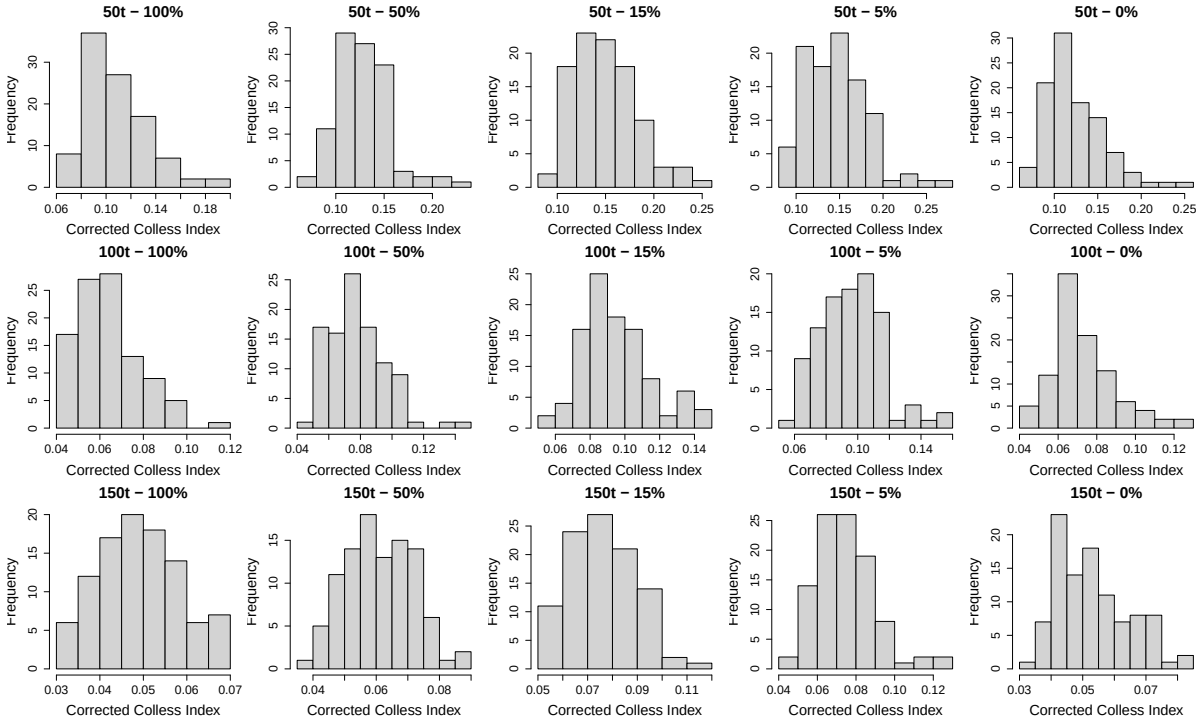

Figure S3: **Main analyses tree symmetry distributions.** Distribution of tree symmetry values measured using the corrected Colless index (Heard, 1992).

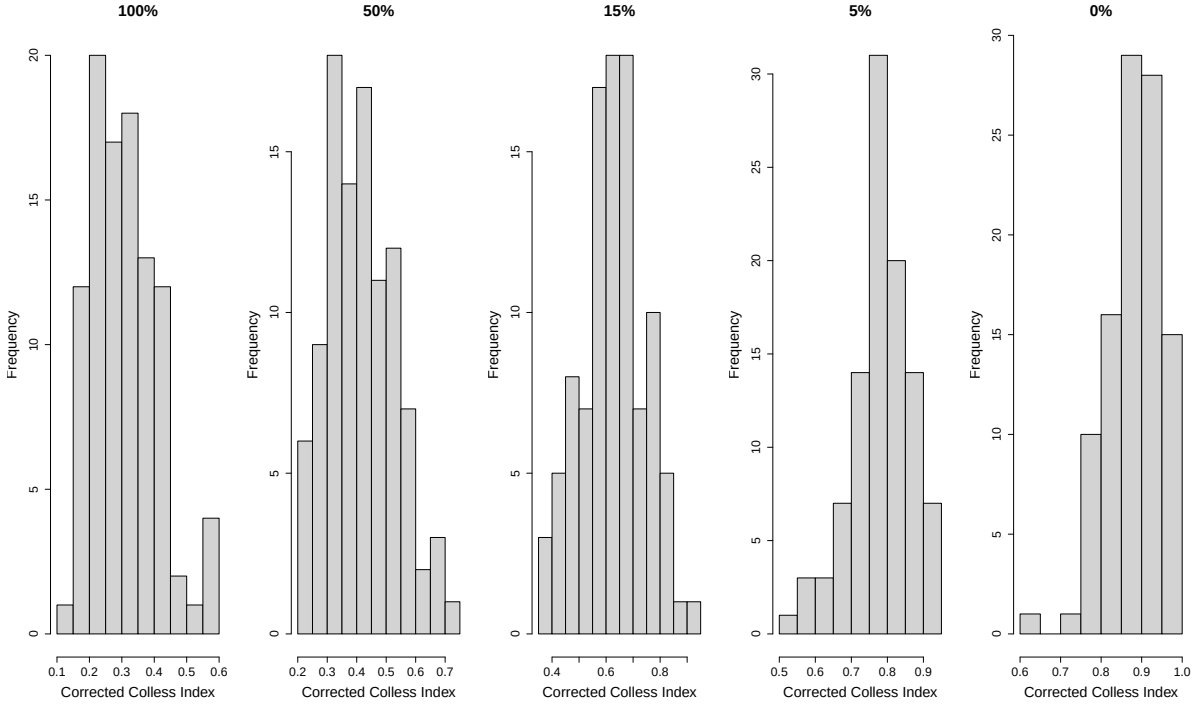

Figure S4: **Asymmetrical trees sensitivity test Colless index distributions.** Distribution of tree symmetry values measured using the corrected Colless index (Heard, 1992).

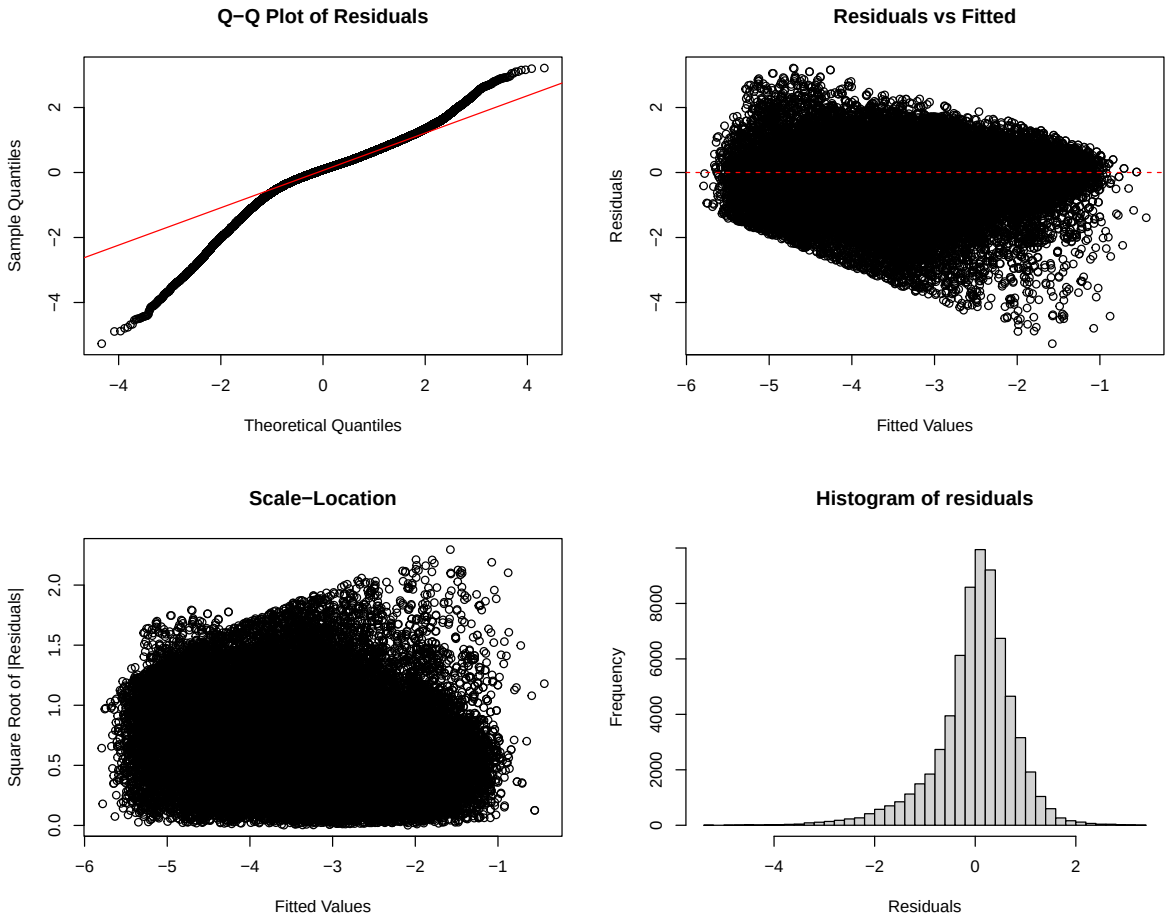

Figure S5: **Diagnostic plots for the continuous trait linear mixed-effects model (aggregated)**.

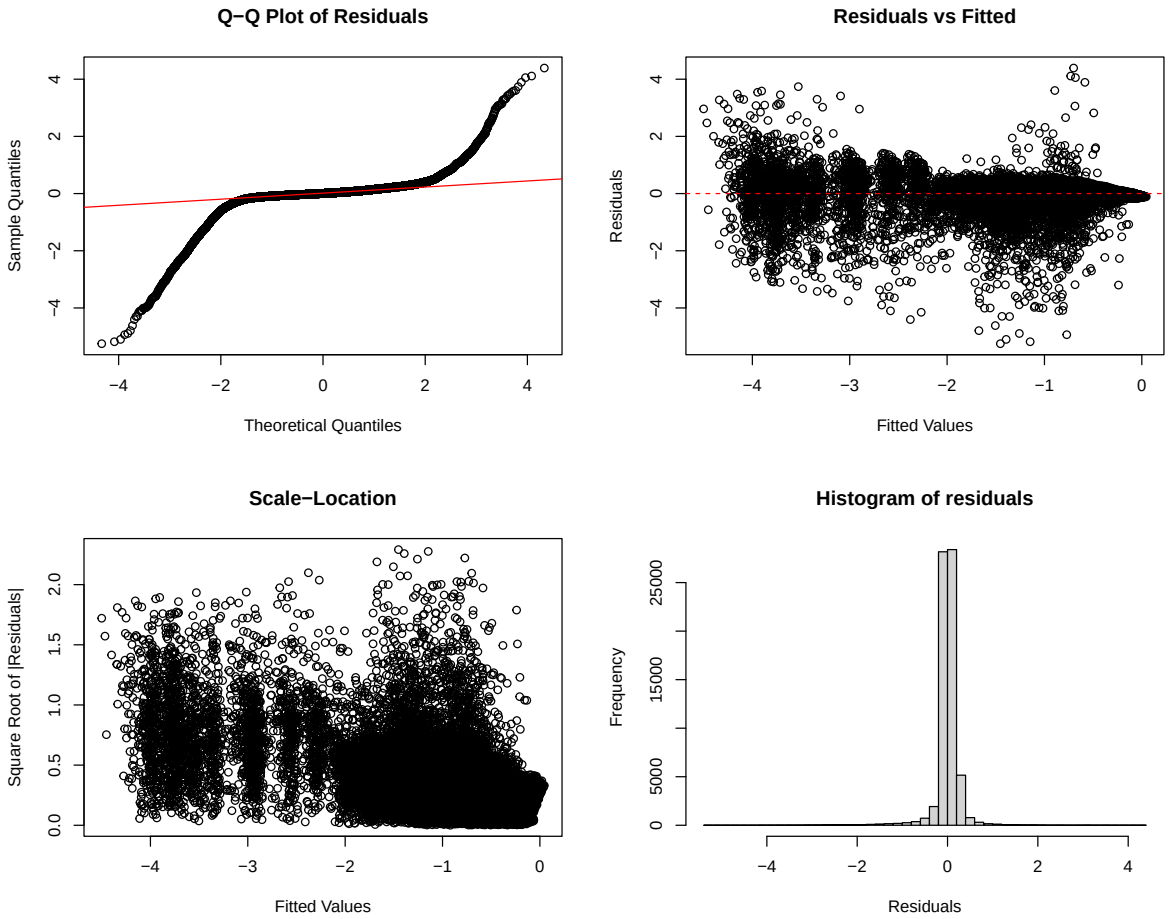

Figure S6: **Diagnostic plots for the discrete trait linear mixed-effects model (aggregated)**

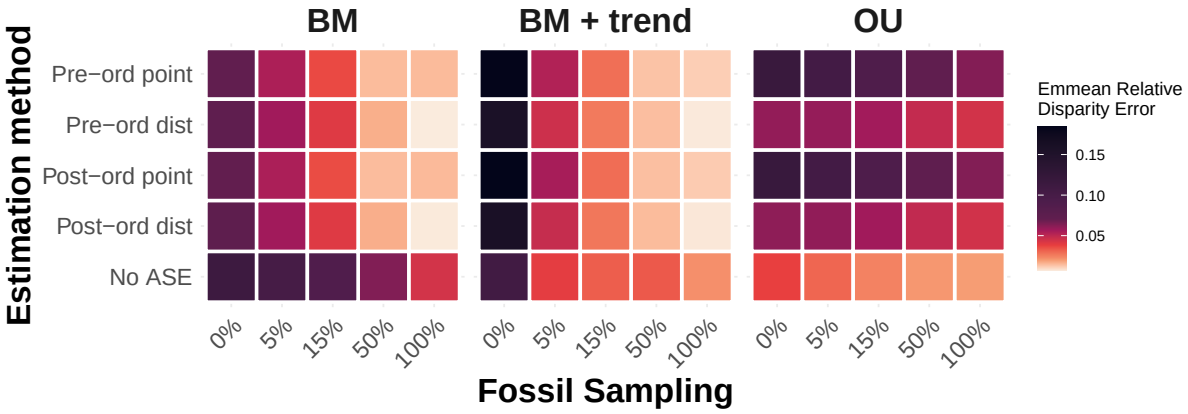

Figure S7: **Weighted LMM continuous heatmap**

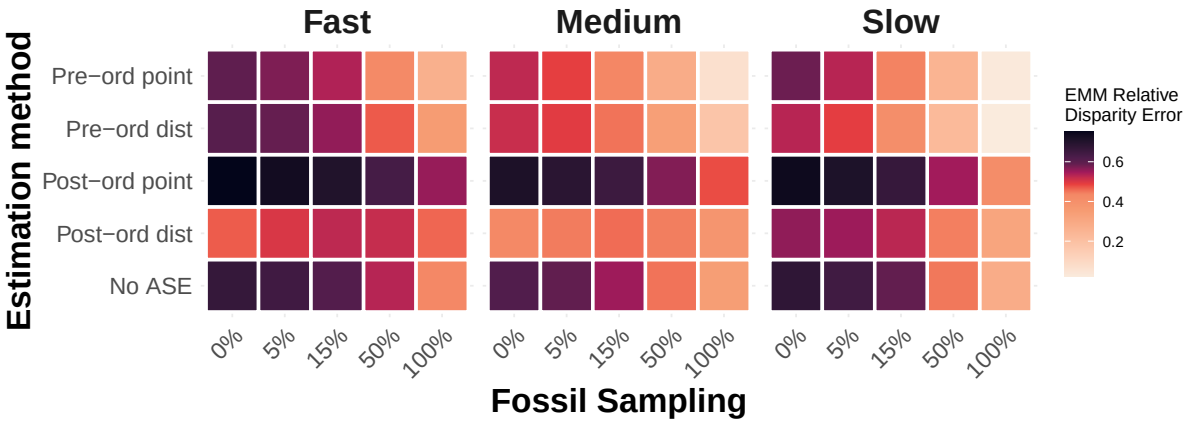

Figure S8: **Weighted LMM discrete heatmap**

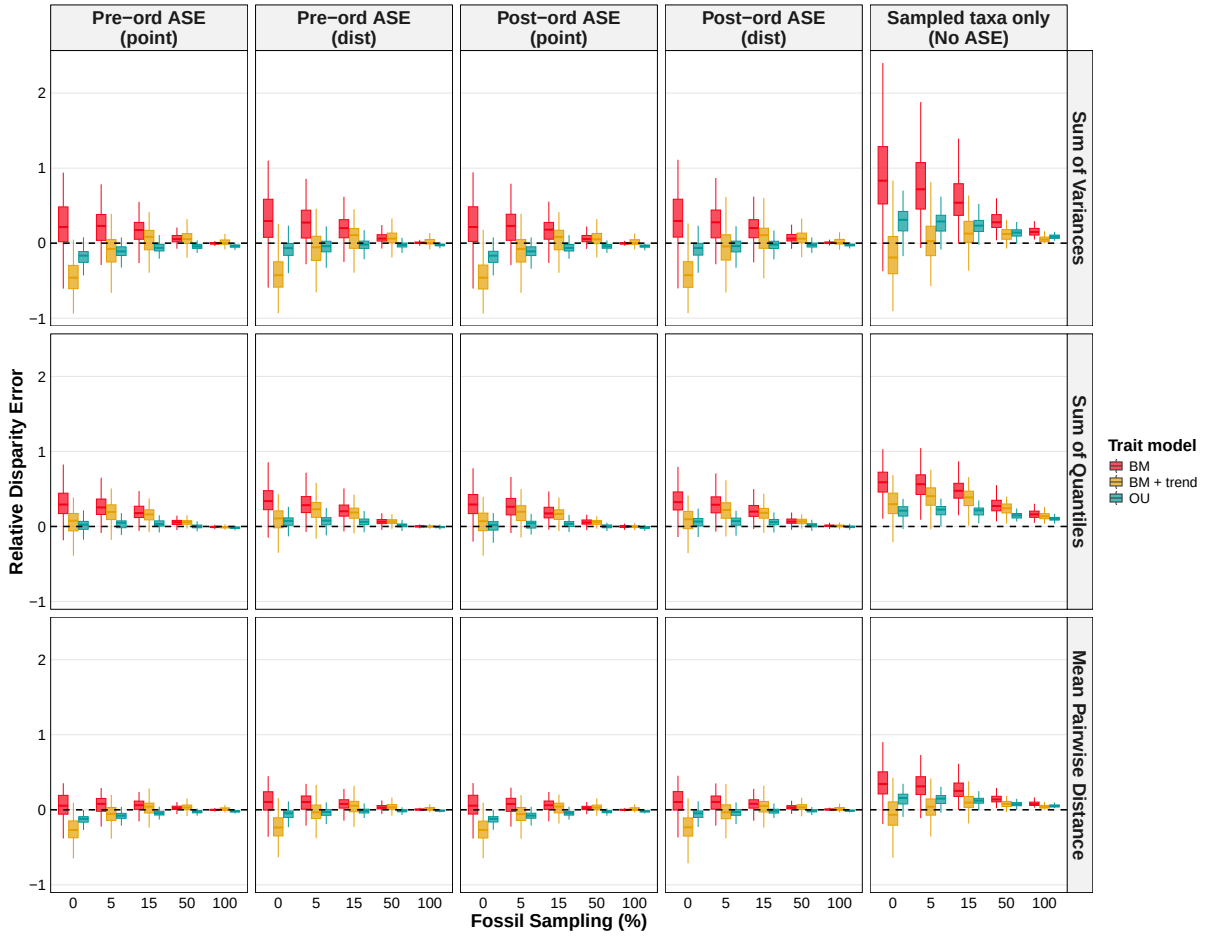

Figure S9: **Asymmetrical tree continuous disparity error**. This is for the tip-only fossil sampling scheme.

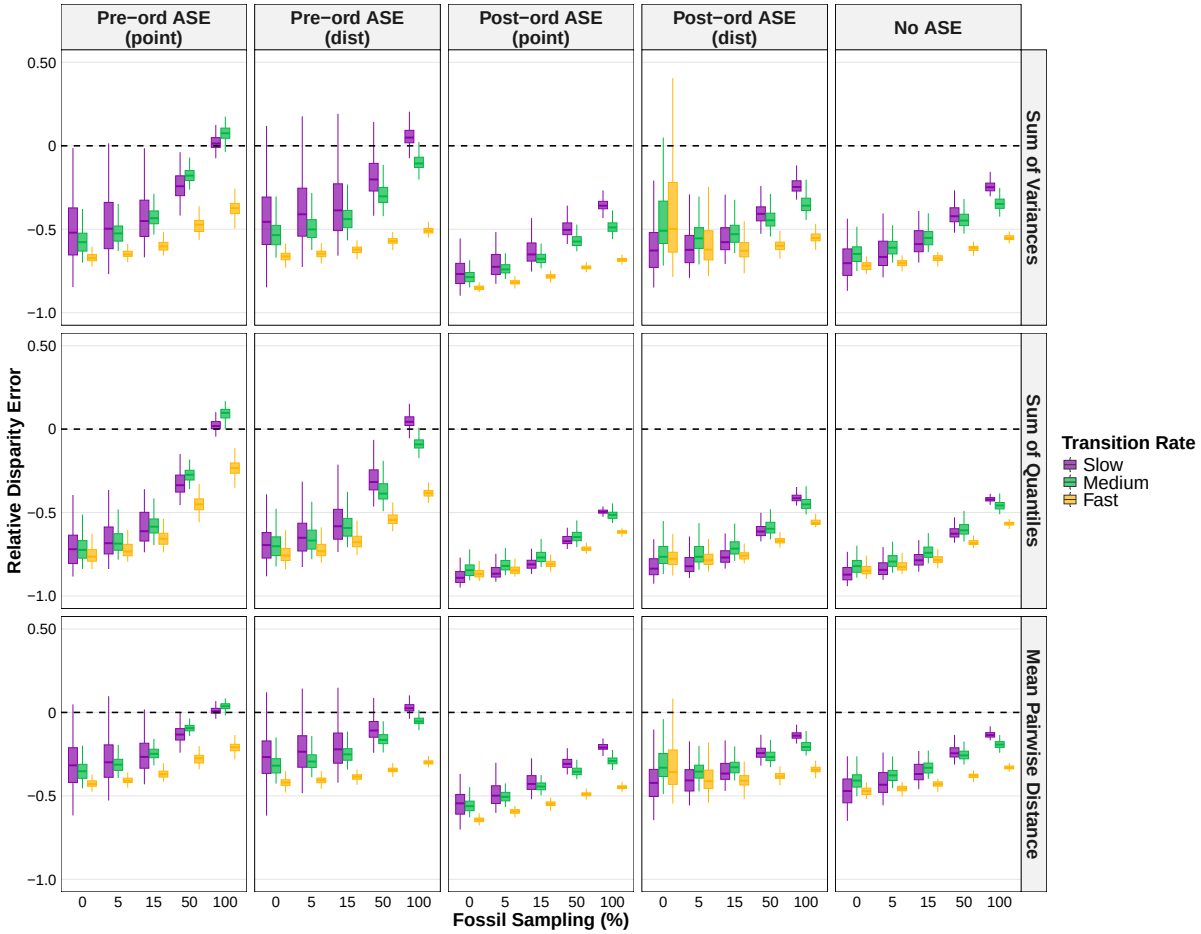

Figure S10: **Asymmetrical tree discrete disparity error**. This is for the tip-only fossil sampling scheme.

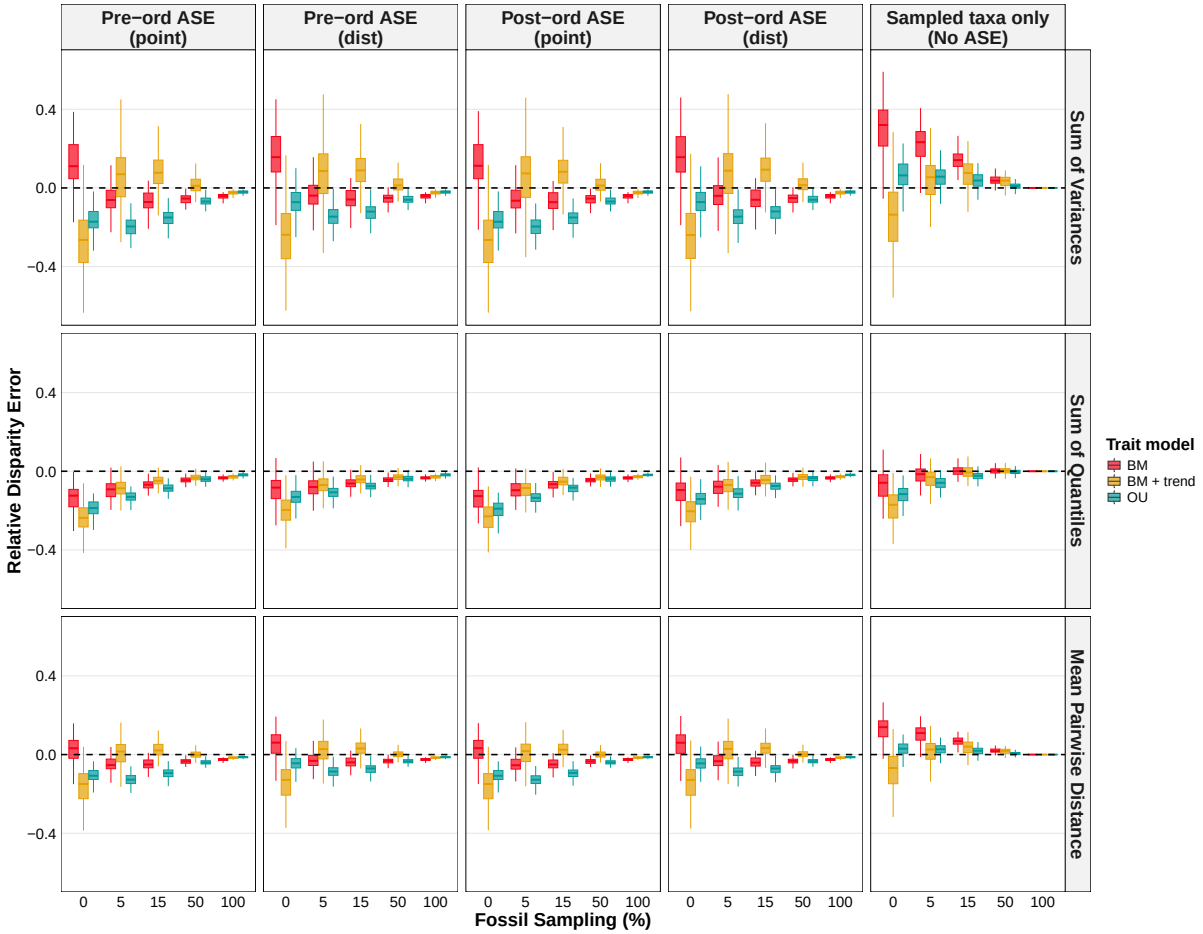

Figure S11: **Continuous disparity error across estimation methods using the alternative tip+nodes fossil sampling scheme**. Relative disparity error returned from pre-ordination point ASE (ancestral state estimation), pre-ordination distribution ASE, post-ordination point ASE, post-ordination distribution ASE and no ASE. These disparity errors were calculated across three transition rates (fast (0.9), medium (0.1) and slow (0.01)), five simulated fossil sampling densities (100%, 50%, 15%, 5% and 0%) and three disparity metrics (Sum of variances, sum of quantiles and mean pairwise distance). Values closest to 0 (dotted horizontal line) were most accurate; negative errors are under-estimates and positive are over-estimates.

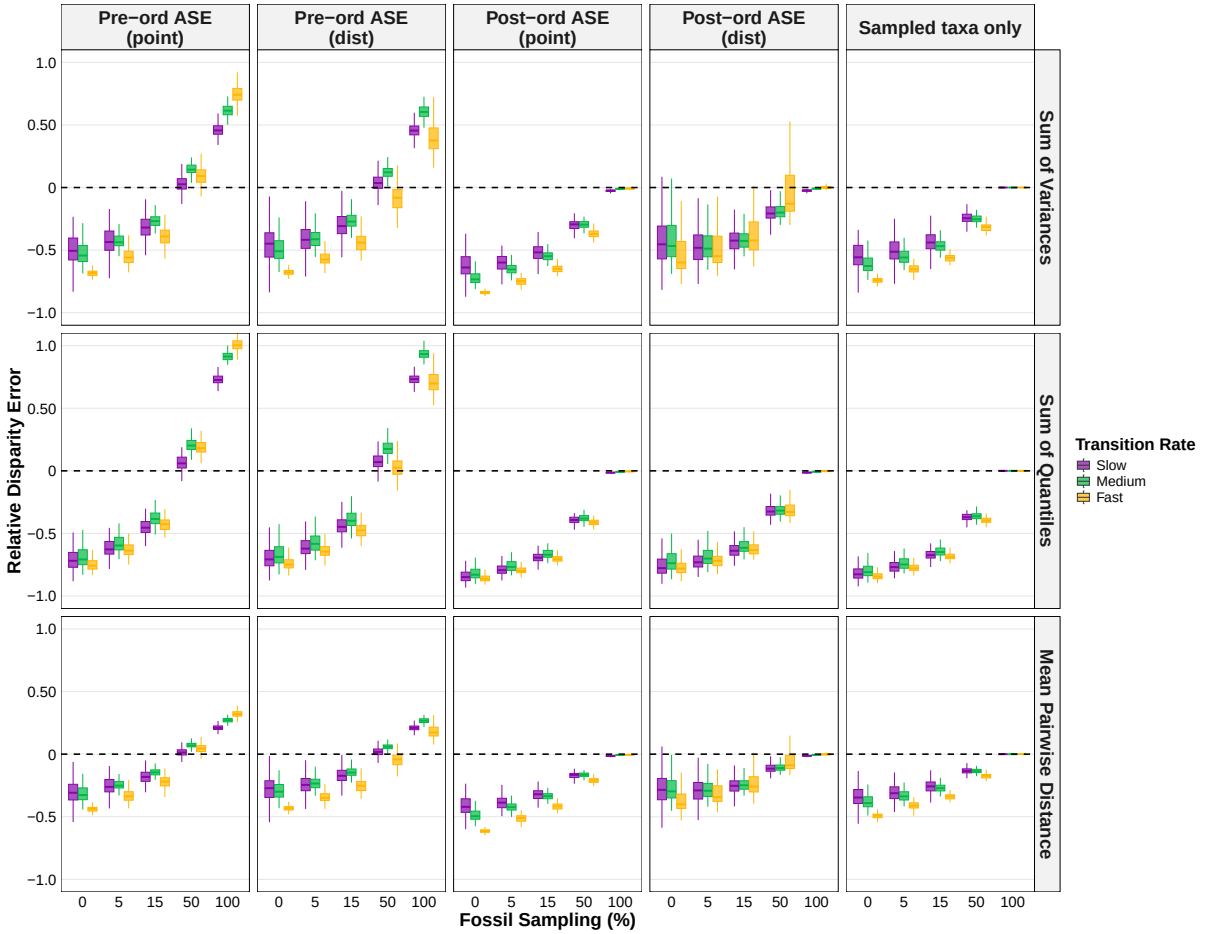

Figure S12: **Discrete disparity error across estimation methods using the alternative tip+nodes fossil sampling scheme**. Relative disparity error returned from pre-ordination point ASE (ancestral state estimation), pre-ordination distribution ASE, post-ordination point ASE, post-ordination distribution ASE and no ASE. These disparity errors were calculated across three transition rates (fast (0.9), medium (0.1) and slow (0.01)), five simulated fossil sampling densities (100%, 50%, 15%, 5% and 0%) and three disparity metrics (Sum of variances, sum of quantiles and mean pairwise distance). Values closest to 0 (dotted horizontal line) were most accurate; negative errors are under-estimates and positive are over-estimates.

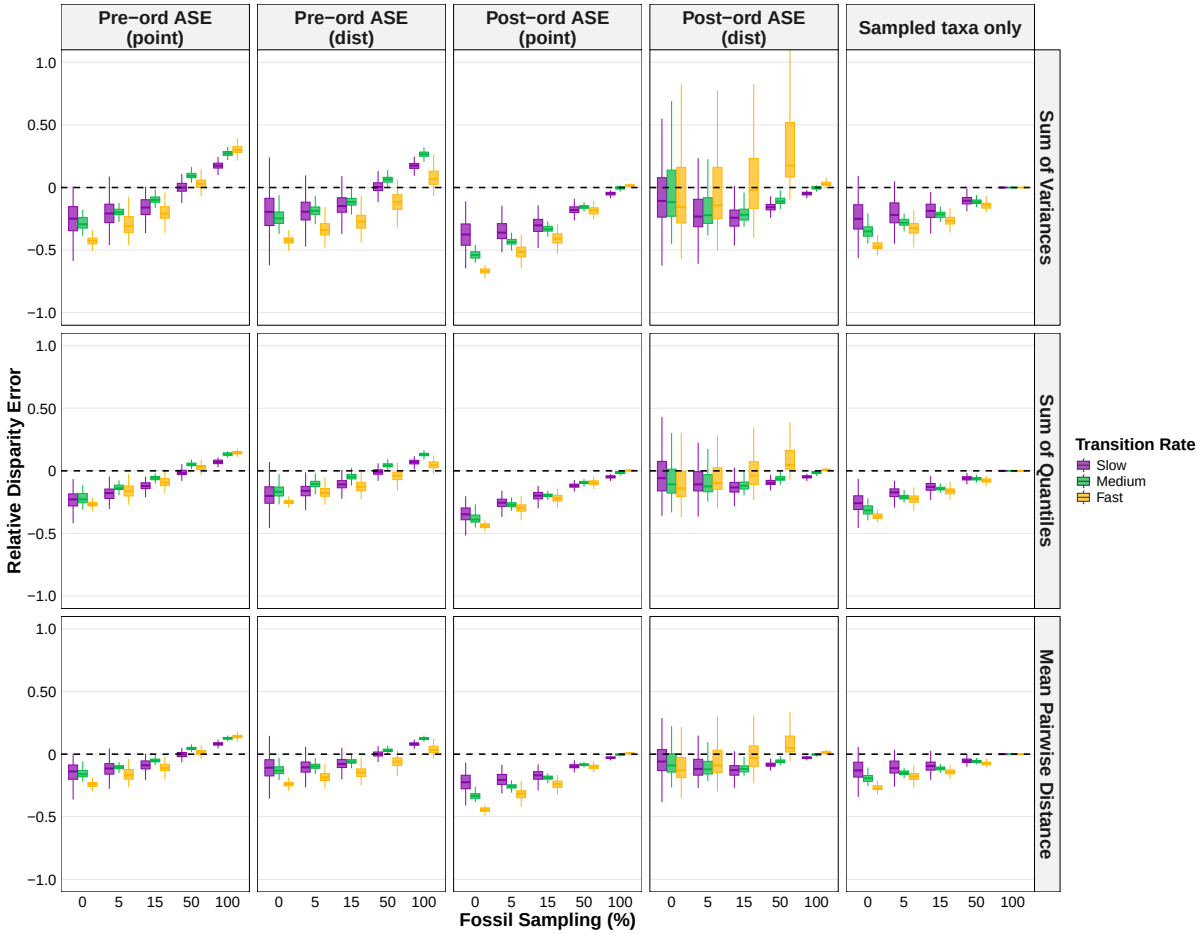

Figure S13: **Discrete disparity error across estimation methods using the alternative tip+nodes fossil sampling scheme**. Relative disparity error returned from pre-ordination point ASE (ancestral state estimation), pre-ordination distribution ASE, post-ordination point ASE, post-ordination distribution ASE and no ASE. These disparity errors were calculated across three transition rates (fast (0.9), medium (0.1) and slow (0.01)), five simulated fossil sampling densities (100%, 50%, 15%, 5% and 0%) and three disparity metrics (Sum of variances, sum of quantiles and mean pairwise distance). Values closest to 0 (dotted horizontal line) were most accurate; negative errors are under-estimates and positive are over-estimates.

References

Stephen B Heard. Patterns in tree balance among cladistic, phenetic, and randomly generated phylogenetic trees. *Evolution*, 46(6):1818–1826, 1992.